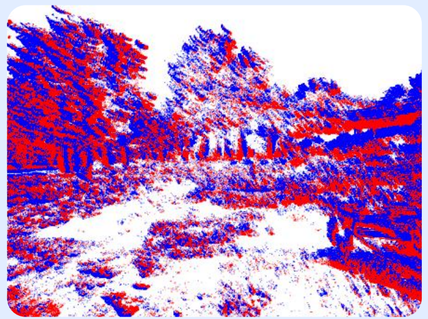
Event Camera



Pre-processing

Deep Network

Tokenizer

Events Former

Prediction Head

$$\begin{bmatrix} {}^{b}\Delta\hat{\mathbf{p}} \\ \hat{\Sigma} \end{bmatrix}$$

IMU

$$\begin{bmatrix} \tilde{\mathbf{a}}_k \\ \tilde{\omega}_k \end{bmatrix}$$

Stochastic Cloning EKF

IMU Model Propagation

Update

$$\begin{bmatrix} {}^{w}_{b}\hat{\mathbf{R}} \\ {}^{w}\hat{\mathbf{v}} \\ {}^{w}\hat{\mathbf{p}} \\ \hat{\mathbf{b}}_g \\ \hat{\mathbf{b}}_a \end{bmatrix}$$